Product Rule

Fact (Product Rule)

Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure.

Example

A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Product Rule

Example

There are 32 computers in a data center in the cloud. Each of these computers has 24 ports. How many different computer ports are there in this data center?

Example

How many different bit strings of length seven are there?

Product Rule

Example

How many functions are there from a set with m elements to a set with n elements?

Example

How many 1-1 functions are there from a set with m elements to a set with n elements?

Solution:

The answer is $n \cdot (n-1) \cdot (n-1) \cdots (n-m+1) = \frac{n!}{(n-m)!}$. We remark that if $n < m$ some of these terms will be 0, hence there will be no 1-1 function from A to B. □

Product Rule

Example

Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$.

Solution:

Let $|S| = n$. Thus, S is a finite set. We know that there is a 1-1 correspondence between the power set of S and the set of functions from S to $\{0, 1\}$. It should be clear that there is a 1-1 correspondence between thhe set of functions from S to $\{0, 1\}$ and the set of bit strings of length $|S|$. Now we know from a previous example that the number of bit strings of length $|S$ is $2^{|S|}$. Therefore,

$$|2^S| = 2^{|S|}.$$

□

Product Rule

Example

Chalk comes in three different lengths, eight different colors, and four different diameters. How many different kinds of chalk are there?

Solution:

The answer is

$$3 \times 8 \times 4 = 96.$$

□

Product Rule

Example

Determine the number of positive integers that divides the number

$$2^4 \times 3^3 \times 5^4 \times 11^2.$$

Solution:

Let $N = 2^4 \times 3^3 \times 5^4 \times 11^2$. If n is a positive integer that divides N, then it has only prime divisors from the list $\{2, 3, 5, 11\}$. Moreover, it should be of the form

$$n = 2^a \times 3^b \times 5^c \times 11^d$$

where

$$0 \leq a \leq 4; \quad 0 \leq b \leq 3, \quad 0 \leq c \leq 4 \text{ and } 0 \leq d \leq 2.$$

Therefore we have 5 choices for a, 4 choices for b, 5 choices for c and 3 choices for d. Hence, we have $5 \times 4 \times 5 \times 5 \times 3 = 300$ choices.

Product Rule

Example

How many two digit numbers have distinct and nonzero digits?

Solution:

Any two digit number is of the form ab where a and b are integers in $\{0, 1, ..., 9\}$. There are 9 possibilities for a non-zero digit. We want them to be distinct. Therefore, we have

$$9 \times 8 = 72$$

possibilities.

There is another method: The number of two digit numbers $\{10, 11, ..., 99\}$ is 90. Nine of them have zeros: 10,20,...,90. Also nine of them have identical digits: 11,22,33,...,99. Observe that they don't have a common element. Therefore, the number of two digit numbers with different and nonzero digits is

$$90 - 9 - 9 = 72.$$

Product Rule

Fact (Product Rule; General Form)

$$|A_1 \times A_2 \times \cdots A_m| = |A_1| \cdot |A_2| \cdots |A_m|.$$

Sum Rule

Fact (Sum Rule)

Suppose that a set is partitioned into pairwise disjoint parts $S_1, S_2, ..., S_m$. The number of objects in S can be determined by finding the number of objects in each of the parts, adding the numbers obtained:

$$|S| = |S_1| + |S_2| + \cdots + |S_m|.$$

Sum Rule

Example

Suppose that either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?

Solution:

□

Sum Rule

Example

A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

Solution:

□

Subtraction Rule

Fact (Subtraction Rule)

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Subtraction Rule

Example

How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution:

Let A be the set of all strings of length 8 which starts with 1, B be the set of strings of length 8 which ends in 00. Then, we have $|A| = 2^7$, $|B| = 2^6$ and $|A \cap B| = 2^5$. Therefore, we have $|A \cup B| = 2^7 + 2^6 - 2^5 = 128 + 64 - 32 = 160$. □

Subtraction Rule

Example

A computer company receives 350 applications from college graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

Solution:

Let C be the set of applicants with major in computer science, B be the set of applicants with major in business. Then we have $|C| = 220$. $|B| = 147$ and $|B \cap C| = 51$. Therefore, we have $|B \cup C| = 220 + 147 - 51 = 316$. We also have $|U| = 350$. Therefore, $|(B \cup C)^c| = 350 - 316 = 34$. □

Division Rule

Fact (Division Rule)

There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

Division Rule

Example

Suppose that an automated system has been developed that counts the legs of cows in a pasture. Suppose that this system has determined that in a farmer's pasture there are exactly 572 legs. How many cows are there is this pasture, assuming that each cow has four legs and that there are no other animals present?

Solution:

□

Division Rule

Example

How many different ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left neighbor and the same right neighbor?

Solution:

Fix one of the people. Then the others will be as if they were on a line. Therefore we have 3! ways. □

Basic Counting

A great many counting problems can be classified as one of the following types:

1. Count the number of *ordered* arrangements or *ordered* selections of objects
   - 1.1 without repeating any object,
   - 1.2 with repetition of objects permitted (but perhaps limited).
2. Count the number of *unordered* arrangements or *unordered* selections of objects
   - 2.1 without repeating any object,
   - 2.2 with repetition of objects permitted (but perhaps limited).

Basic Counting

Instead of distinguishing between nonrepetition and repetition of objects, it is sometimes more convenient to distinguish between selections from a set and a multiset. A **multiset** is like a set except that its members need not be distinct. For example, we might have a multiset M with three a's, one b, two e's, and four d's, that is, 10 elements of 4 different types: 3 of type a, 1 of type b, 2 of type e, and 4 of type d. We shall usually indicate a multiset by specifying the number of times different types of elements occur in it. Thus, M shall be denoted by $\{3 \cdot a, 1 \cdot b, 2 \cdot e, 4 \cdot d\}$. The numbers 3,1,2, and 4 are the repetition numbers of the multiset M. A set is a multiset that has all repetition numbers equal to 1. To include the listed case (b) when there is no limit on the number of times an object of each type can occur (except for that imposed by the size of the arrangement), we allow infinite repetition numbers.

Basic Counting

Thus, a multiset in which a and c each have an infinite repetition number and band d have repetition numbers 2 and 4, respectively, is denoted by $\{\infty \cdot a, 2 \cdot b, \infty \cdot c, 4 \cdot d\}$. Arrangements or selections in (1) in which order is taken into consideration are generally called **permutations**, whereas arrangements or selections in (2) in which order is irrelevant are generally called **combinations**.

Basic Counting

Example

How many odd numbers between 1000 and 9999 have distinct digits?

Solution:

□

Basic Counting

Example

How many integers between 0 and 10,000 have only one digit equal to 5?

Solution:

S_i is the number of i-digit such numbers.
Then $S_1 = 1$, $S_2 = 17$, $S_3 = 225$, $S_4 = 2673$.
Then $S = S_1 + S_2 + S_3 + S_4 = 2916$. □

Basic Counting

Example

How many different five-digit numbers can be constructed out of the digits 1, 1, 1, 3, 8?

Solution:

We have $\frac{5!}{3!}$ such numbers. (Division Rule) □

Example

How many functions are there from the set $\{1, 2, \ldots, n\}$, where n is a positive integer, to the set $\{0, 1\}$.

1. that are one-to-one?
2. that assign 0 to both 1 and n?
3. that assign 1 to exactly one of the positive integers less than n?

Solution:

□

1. If $n \leq 2$ we have two 1-1 functions. If $n > 2$ there is none.
2. The values of 1 and n are fixed. Therefore we have 2^{n-2} such functions.
3. n has two choices. The rest is determined by the place of unique 1. Therefore we have $2(n-1)$ such functions.

Example

Show that a natural number $n \geq 1$ has an odd number of divisors (including 1 and itself) if and only if $\sqrt{n}$ is an integer.

Solution:

Write $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$. The hypothesis says that $\prod(a_i + 1)$ is odd. Therefore, all of a_i are even. Then $\prod p_i^{a_i/2} = \sqrt{n}$ which is an integer. The converse is similar to the proof that $\sqrt{2}$ is irrational. (Why?) □

Basic Counting

Recall that there are $\frac{n!}{(n-k)!}$ 1-1 functions $f : \{1, 2, ..., k\} \to S$ where S is a set with n elements. This means that there are $P(n, k) = \frac{n!}{(n-k)!}$ ordered stings of elements of S of length k. These are called k-**permutations** of S. Any k-element subset of S has $k!$ different orderings (permutations). Therefore, we have

$$\frac{\frac{n!}{(n-k)!}}{k!}$$

k-element subsets of S. They are called the k-**combinations** of S. Hence, we have

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

k-combinations of S.

Example

What is the number of ways to order the 26 letters of the alphabet so that no two of the vowels a, e, i, o, and u occur consecutively?

Solution:

There are 21 consonants, and so 21! permutations of the consonants. Since we cannot have two consecutive vowels in our final arrangement, the vowels must be in 5 of the 22 spaces before, between, and after the consonants. Our second task is to put the vowels in these places. There are 22 places for the a, then 21 for the e, 20 for the i, 19 for the 0, and 18 for the u. That is, the second task can be accomplished in $P(22, 5) = \frac{22!}{17!}$ ways. By the multiplication principle, we determine that the number of ordered ar- rangements of the letters of the alphabet with no two vowels consecutive is $21! \times \frac{22!}{17!}$. □

Basic Counting

Example

How many seven-digit numbers are there such that the digits are distinct integers taken from $\{1, 2, ..., 9\}$ and such that the digits 5 and 6 do not appear consecutively in either order?

Solution:

Let T be the set of 7-digit numbers formed by using distinct digits from $\{1, 2, 3, ..., 9\}$. Then $|T| = P(9, 7) = \frac{9!}{2!} = 181,440$. Let S be the elements of T in which 5 and 6 do not occur consequently. Then $T - S$ consists of numbers where 5 and 6 occur consequently. Then

$$|S^c| = 2 \times 6 \times P(7, 5) = 30,240.$$

Therefore,

$$|S| = 181,440 - 30,240 = 151,200.$$

□

Basic Counting

Ordinary permutations are orderings on the line. If we consider orderings on a circle, we speak of circular permutations.

Theorem (Circular r-permutations)

The number of circular r-permutations of a set of n elements is given by

$$\frac{P(n,r)}{r} = \frac{n!}{r \cdot (n-r)!}.$$

Corollary

The number of circular n-permutaions is then

$$\frac{P(n,n)}{n} = \frac{n!}{n \cdot 0!} = (n-1)!$$

Basic Counting

Example

Ten people, including two who do not wish to sit next to one another, are to be seated at a round table. How many circular seating arrangements are there?

Solution:

First seat P_1 at the "head" of the table. Then P_2 cannot be on either side of P_1. There are 8 choices for the person on P_1's left, 7 choices for the person on P_1's right, and the remaining seats can be filled in 7! ways. Thus, the number of seating arrangements in which P_1 and P_2 are not together is

$$8 \times 7 \times 7! = 7 \times 8!.$$

□

Example

Twenty-five points are chosen in the plane so that no three of them are collinear. How many straight lines do they determine? How many triangles do they determine?

Solution:

The number of lines is $\binom{25}{2}$.

There are $\binom{25}{3}$ triangles. □

Example

How many eight-letter words can be constructed by using the 26 letters of the alphabet if each word contains three, four, or five vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

Solution:

$$N_3 = \binom{8}{3} 5^3 21^5.$$

$$N_4 = \binom{8}{4} 5^4 21^4.$$

$$N_5 = \binom{8}{5} 5^5 21^3.$$

Hence, $N = N_1 + N_2 + N_3$. □

Binomial Formula

Let us consider the product $(x+y)^n$. Each term will be of the form $x^p y^q$ where $p+q=n$. When we calculate each term, we can consider the set of words of length n written with the letters x and y. We can write a n-string with k times xs in $\binom{n}{k}$ many ways. Therefore, when we multiply $(x+y)^n = (x+y)(x+y)\cdots(x+y)$, we have $\binom{n}{k}$ many $x^k y^{n-k}$, for each k. Therefore, we have

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Two Special Cases

Corollary

We put $x = y = 1$. Then we have

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Therefore, a set with n elements has 2^n subsets. This is the third proof we have for this very important result.

Two Special Cases

Corollary

Now put $x = 1$ and $y = -1$. Then we have

$$\sum_{k=0}^{n} \binom{n}{k}(-1)^k = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots (-1)^n \binom{n}{n} = 0.$$

Let S be a set with n elements. Let us call a subset $A \subseteq S$ even if it has an even number of elements. Similarly odd subsets are defined. This corollary says that the number of even subsets of S and the number of odd subsets of S are equal.

Binomial Formula

There are many properties of these coefficients: For example, one can show very easily that

$$\binom{n}{k} = \binom{n}{n-k}.$$

Theorem (Pascal's Identity)

Let n and k be positive integers with $k \leq n$. Then we have

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof.

□

Binomial Formula

Example

Consider $X = \{A \subset [n] : |A| = 2\}$.
For each $i \leq n$ let $X_i = \{A \in X : \max(A) = i\}$.
Then we have $|X_1| = 0$, $|X_2| = 1$, $|X_3| = 2$,...,$|X_i| = i - 1$,...,$|X_n| = n - 1$.
We also have $X = X_1 \cup X_2 \cup X_3 \cup ... \cup X_n$ (disjoint). Therefore, we have $|X| = \sum |X_i|$, hence

$$0 + 1 + 2 + \cdots + (n-1) = \binom{n}{2} = \frac{n.(n-1)}{2}.$$

Permutations of Multisets

If S is a multiset, an r-permutation of S is an ordered arrangement of r of the objects of S. If the total number of objects of S is n (counting repetitions), then an n-permutation of S will also be called a permutation of S.

Theorem

Let S .be a multiset with objects of k different types, where each object has an infinite repetition number. Then the number of r-permutations of S is k^r.

Permutations of Multisets

Example

What is the number of ternary numerals with at most four digits?

Permutations of Multisets

Theorem

Let S be a multiset with objects of k different types with finite repetition numbers $n_1, n_2, ..., n_k$, respectively. Let the size of S be $n = n_1 + n_2 + ... + n_k$. Then the number of permutations of S equals

$$\frac{n!}{n_1! \cdot n_2! \cdots n_k!}.$$

Idea of proof:

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-...n_{k-1}}{n_k} = \frac{n!}{n_1!n_2!...n_k!}.$$

□

Permutations of Multisets

Example

The number of permutations of the letters in the word MISSISSIPPI is

$$\frac{11!}{1!4!4!2!}.$$

Solution:

Here our multiset is $\{1 \cdot M, 4 \cdot I, 4 \cdot S, 2 \cdot P\}$. The rest follows. □

Permutations of Multisets

Example

Let $S = \{a, b, c, d\}$. Partition S into 2 sets of equal size. Then we have 3 such partitions:

$$\{a, b\}, \{c, d\}; \quad \{a, c\}, \{b, d\}, \quad \{a, d\}, \{b, c\}.$$

Now suppose that the parts are labeled by e.g colors red and blue. Then we have 6 of them:....For example, for the partition $\{a, b\}, \{c, d\}$, we have

$$\{a, b\}, \{c, d\}$$

$$\{a, b\}, \{c, d\}$$

This way we have 6 labeled partitions.

Permutations of Multisets

Theorem

Let n be a positive integer and let $n_1, n_2, ..., n_k$ be positive integers with $n = n_1 + n_2 + ... + n_k$ The number of ways to partition a set of n objects into k labeled boxes in which Box 1 contains n_1 objects, Box 2 contains n_2 objects, ... , Box k contains n_k objects equals

$$\frac{n!}{n_1! \cdot n_2! \cdots n_k!}.$$

If the boxes are not labeled and $n_i = n_j$ for every i, j, then it is equal to

$$\frac{n!}{k! n_1! n_2! ... n_k!}.$$

Permutations of Multisets

Example

Let $S = \{3 \cdot a, 2 \cdot b, 4 \cdot c\}$. Find the number of 8-permutations of S.

Solution:

1. $S_1 = S = \{2 \cdot a, 2 \cdot b, 4 \cdot c\}$. Then we have $N_1 = \frac{8!}{2!2!4!} = 420$
2. $S_2 = S = \{3 \cdot a, 1 \cdot b, 4 \cdot c\}$. Then we have $N_2 = \frac{8!}{3!1!4!} = 280$.
3. $S_3 = S = \{3 \cdot a, 2 \cdot b, 3 \cdot c\}$. Then we have $N_3 = \frac{8!}{3!2!3!} = 560$.

The answer is $N = N_1 + N_2 + N_3$. □

Combinations of Multisets

Example

Let $S = \{2 \cdot a, 1 \cdot b, 3 \cdot c\}$. Then the 3-combinations of S are

$$\{2 \cdot a, 1 \cdot b\}, \quad \{2 \cdot a, 1 \cdot c, \quad \{1 \cdot a, 1 \cdot b, 1 \cdot c\},$$

$$\{1 \cdot a, 2 \cdot c\}, \quad \{1 \cdot b, 2 \cdot c\}, \quad \{3 \cdot c\}.$$

Hence, there are 6 of them.

Combinations of Multisets

Theorem

Let S be a multiset with objects with k different types, each with an infinite repetition number. Then the number of r-combinations of S is given by

$$\binom{r+k-1}{r} = \binom{r+k-1}{k-1}.$$

Idea of proof.

We have $S = \{\infty \cdot a_1, \infty \cdot a_2, ..., \infty \cdot a_n\}$. An r-combination of S is of the form $\{x_1 \cdot a_1, x_2 \cdot a_2, ..., x_k \cdot a_k\}$ where $x_1, x_2, .., x_k$ are non-negative integers with $x_1 + x_2 + ... + x_k = r$. It is clear that there is a 1-1 correspondence between the set of r-combinations of S and the non-negative integer solutions of the equation

$$x_1 + x_2 + ... + x_k = r.$$

Combinations of Multisets

This number is equal to the number of permutations of the multiset

$$T = \{r \cdot 1, (k-1) \cdot *\}.$$

Therefore, we have

$$\binom{r+k-1}{k}$$

permutations. The thorem follows.

Combinations of Multisets

Example

A bakery boasts eight varieties of doughnuts. If a box of doughnuts contains one dozen, how many different options are there for a box of doughnuts?

Solution:

We have

$$\binom{12+8-1}{12} = \binom{19}{12}$$

possibilities. □

Combinations of Multisets

Example

What is the number of nondecreasing sequences of length r whose terms are taken from 1,2, ... , k?

Solution:

The nondecreasing sequences can be obtained by finding r-combinations of the multiset

$$\{\infty \cdot 1, \infty \cdot 2, ..., \infty \cdot k\}$$

and then writing them in nondecreasing order. There there are

$$\binom{r+k-1}{r}$$

such sequences. □

Combinations of Multisets

Example

Let S be the multiset $\{\infty \cdot a, \infty \cdot b, \infty \cdot c, \infty \cdot d\}$ with objects of four types, a, b, c, and d. What is the number of 10-combinations of S that have the property that each of the four types of objects occurs at least once?

Solution:

We are interested in **positive** integer solutions of

$$x_1 + x_2 + x_3 + x_4 = 10.$$

Thus, we are interested in nonnegative integer solutions of

$$y_1 + ty_2 + y_3 + y_4 = 6$$

where $y_i = x_i - 1$ for i =1,2,3,4. Therefore we have $\binom{6+4-1}{6} = \binom{9}{6} = 84$ combinations. □

Combinations of Multisets

Example

What is the number of integral solutions of

$$x_1 + x_2 + x_3 + x_4 = 20$$

where we have

$$x_1 \geq 3, x_2 \geq 1, x_3 \geq 0 \text{ and } x_4 \geq 5?$$

Solution:

Put $y_1 = x_1 - 3, y_2 = x_2 - 1, y_3 = x_3, y_4 = x_4 - 5$. Then we have

$$y_1 + y_2 + y_3 + y_4 = 11.$$

Thus, we have $\binom{11+4-1}{11} = 364$ solutions. □

Binomial Formula

Theorem (Vandermonde's Identity)

m, n, r are positive integers with $r \leq m$ and $r \leq n$. Then we have

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}.$$

Proof.

□

Corollary

Let n be a positive integer. Then we have

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

Pigeonhole Principle

Pigeonhole Principle: Simple Form

Theorem

If $n+1$ objects are distrbuted in n boxes, then at least one of the boxes contains two or more objects.

One can prove this statement by contradiction!

Example

1. Among any 13 people there are 2 who have their birthdays in the same month.
2. There are n married couples. This way we have in total $2n$ people. Among any $n+1$ of then, there is a married couple.

Pigeonhole Principle

Some variants:

1. If n objects are put into n boxes and if no box is empty, then each box contains exactly one object.
2. If n objects are put into n boxes and no box gets more than one object, then every box has an object in it.

Pigeonhole Principle

Another Formulation:

Suppose that X and Y are two finite sets and let $f : X \to Y$ be a function.

1. If $|Y| < |X|$, then f can not be 1-1.
2. If $|X| = |Y|$ and f is onto, then it is also 1-1.
3. If $|X| = |Y|$ and f is 1-1, then it is onto.

Pigeonhole Principle

Example

From the integers 1,2, ... ,200, we choose 101 integers. Show that, among the integers chosen, there are two such that one of them is divisible by the other.

Idea.

Every integer is of the form $2^k a$ where k is a nonnegative integer and a is an odd integer. □

Pigeonhole Principle

Theorem

If $mn + 1$ pigeons are placed into n pigeonholes, then there is a pigeonhole with at least $m + 1$ pigeons.

Theorem

Let $m_1, m_2, ..., m_n$ be positive integers. If $m_1 + m_2 + ... + m_n - n + 1$ pigeons are placed into n pigeonholes, then for some $1 \leq i \leq n$ such that ith pigeonhole contains at least m_i pigeons.

Again, one can prove this statemnt by contradiction!.

Pigeonhole Principle

Example

There is a nonzero multiple of 6 whose digits are all zeros or ones.

Idea.

Consider the any 6 different numbers consisting of only ones. Write $a_1 < a_2, ... < a_6$. If one of them is divisible by 6, we are done. Suppose not. Let r_1 be the remainder when we divide a_1 by 6, r_2 be the remaninder when we divide a_2 by 6,... Then $r_1, r_2, ..., r_6 \in \{1, 2, 3, 4, 5\}$. By the pigeonhole principle there are $1 \leq i < j \leq 6$ such that $r_i = r_j$. Now it is easy to see that $a_j - a_i$ has digits equal to either 0 or 1, and $a_j - a_i$ is divisible by 6. □

Pigeonhole Principle

Example

Let S be a set consisting of $n \geq 2$ integers. Then there is a subset A of S such that the sum of elements of A is a multiple of n.

Solution:

Let $S_1 = \{a_1\}$, $S_2 = \{a_1, a_2\}$,...,$S_n = S$. Let $s_k = a_1 + a_2, .. + a_k$ be the sum of elements in S_k. If there is some k such that $n|s_k$, we have nothing to prove. If not, by the pigeon principle, there are $k < l$ such that s_k and s_l have the same remainders when divided by n. (Look at the previous example!) This means that $s_l - s_k$ is divisible by n. But $s_l - s_k$ is nothing but the sum of elements in $A - S_l - S_k \subset S$. □